Gyroscope-unit sensitivity

Revision excerpt: Methods replacement and Results addition

Methods: unit handling

The README distributed with the CERTH dataset [12] specifies angular velocity in radians per second, whereas the CSV gyroscope column names contain degree-per-second suffixes. The primary analysis followed the README and supplied the released numeric values directly to the AHRS 0.4.0 Madgwick filter, which expects radians per second [13]. A sensitivity analysis evaluated the alternative CSV-header interpretation by multiplying the same gyroscope values by $\pi/180$ before orientation estimation. Both scenarios used 50 Hz IMU updates, filter gain 0.033, and no magnetometer input. Initialization, quaternion normalization and sign continuity, participant groups, evaluation endpoints, model architectures, seeds, and training and validation-selection rules were unchanged. Each alternative full-HTM backbone was trained afresh. The primary convention was determined from the dataset README before comparing performance.

Results

Sensitivity to gyroscope-unit interpretation

Recomputing the README-convention quaternion features reproduced all archived float32 arrays exactly. Under the alternative interpretation, the same source values were treated as degrees per second and converted to radians per second. Let $\mathbf{q}_t^{(R)}$ and $\mathbf{q}_t^{(D)}$ denote the unit quaternions estimated under the README and converted CSV interpretations, respectively. The orientation difference in radians was calculated as $2\arccos(|(\mathbf{q}_t^{(R)})^\top \mathbf{q}_t^{(D)}|)$, with numerical clipping of the absolute inner product to $[0, 1]$. Differences were converted to degrees for reporting. At the 4,636 held-out endpoints, this difference had a median of 35.04° and a 95th percentile of 177.38°. These values describe differences between two orientation estimates; orientation ground truth was unavailable.

Table 1 compares the archived primary fits with models refitted under the alternative interpretation using the same protocol and seeds. Validation selection was repeated within each interpretation. The elapsed-time-only control, which does not use sensor features, produced identical predictions under both scenarios.

Table 1: Sensitivity of test macro-F1 to the interpretation of released gyroscope values. Entries are mean ± sample SD across three model seeds (7, 42, and 2026) on one fixed participant split. The difference is the converted deg/s scenario minus the README rad/s scenario, in percentage points.

Model	README rad/s (%)	CSV deg/s (%)	Difference (pp)
Full HTM	23.58 ± 0.18	14.47 ± 0.98	−9.11
Logistic regression	42.20 ± 0.00	16.75 ± 0.00	−25.45
CNN	75.67 ± 1.01	36.65 ± 0.70	−39.02
LSTM	74.47 ± 6.55	28.16 ± 5.69	−46.31
Transformer	89.29 ± 0.18	82.27 ± 1.81	−7.02
Elapsed time only	79.86 ± 0.26	79.86 ± 0.26	+0.00

The results demonstrate sensitivity to the unit interpretation. Classification performance does not establish the true physical unit, and the conflicting source metadata remain a limitation pending clarification from the dataset creators. The primary results retain the explicit README convention. This sensitivity comparison does not validate orientation accuracy or generalization beyond the fixed participant split.

References

[12] Anagnostis, A.; Benos, L.; Tsaopoulos, D.; Tagarakis, A.; Tsolakis, N.; Bochtis, D. Human activity recognition through recurrent neural networks for human–robot interaction in agriculture. *Appl. Sci.* **2021**, *11*, 2188. https://doi.org/10.3390/app11052188.

[13] AHRS Documentation. Madgwick Orientation Filter (AHRS 0.4.0). Available online: https://ahrs.readthedocs.io/en/latest/filters/madgwick.html (accessed on 16 October 2026).